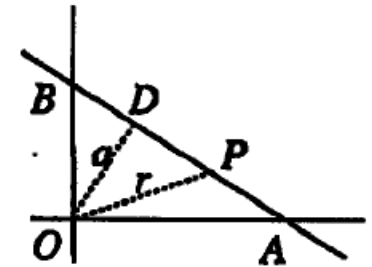


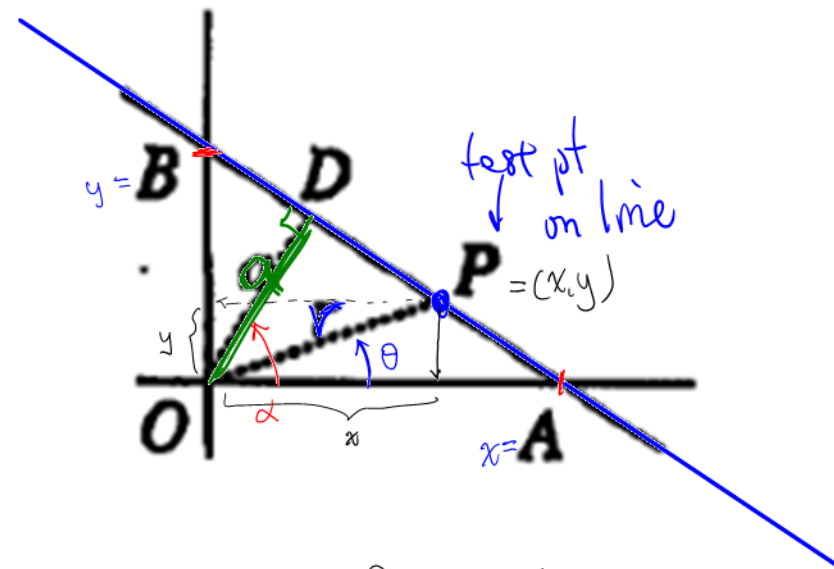
MA1521 L1 Week 03 Mon 20260824

- (i) Find the polar equation for the line intersecting the positive x - and y -axes respectively at A and B , and having perpendicular distance a from the origin.

(Let $\alpha = \angle DOA$; use the right triangle DOP to get the equation connecting r, θ, α and a .



- (ii) Convert your polar equation to the usual rectangular equation involving A and B , by using trigonometry.



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$A \cos \alpha = a$$

$$B \cos \left(\frac{\pi}{2} - \alpha \right) = a$$

$\sin \alpha$

(i) Angle $\angle POD = \alpha - \theta$

Egn is $r \cos(\alpha - \theta) = a$

(ii) "Add formula for cos" gives

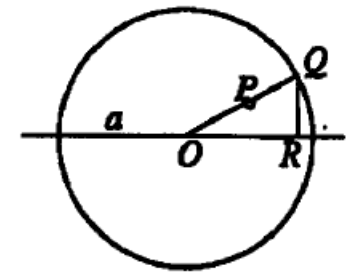
$$r \cdot \cos \alpha \cos \theta + r \sin \alpha \sin \theta = a$$

i.e. $x \cdot \cos \alpha + y \cdot \sin \alpha = a$

i.e. $x \frac{a}{A} + y \frac{a}{B} = a$

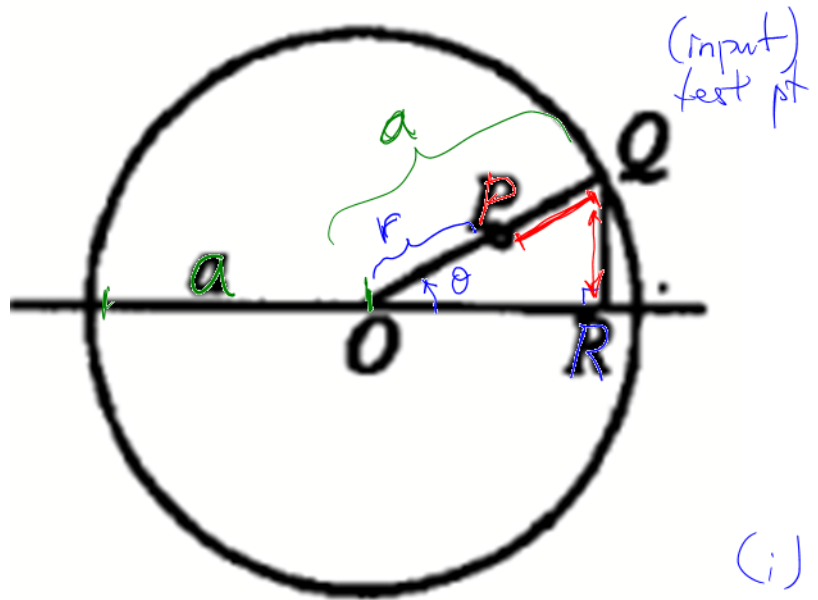
i.e. $\frac{x}{A} + \frac{y}{B} = 1$

In the accompanying figure, the point Q moves around the circle of radius a centered at the origin; QR is a perpendicular to the x -axis. P is a point on ray OQ such that $|QP| = |QR|$: P is the point inside the circle in the first two quadrants, but outside the circle in the last two quadrants.



- (i) Sketch the locus of P ; the locus is called a *cardioid*
- (ii) find the polar equation of this locus.

If P has polar coords r, θ
 then $|\vec{OP}| = r$
 and $\angle PO(+ve\ x\text{-axis}) = \theta$



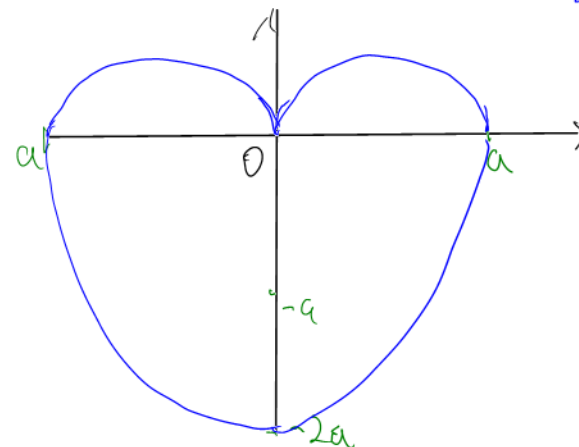
(ii) Relation $|QP| = |QR|$

means $a - r = a \sin \theta$

$r = a(1 - \sin \theta)$

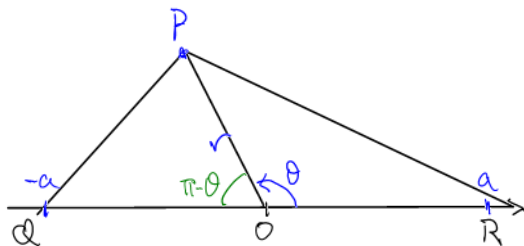
(check against rule that P is the point inside the circle in the first two quadrants, outside the circle in the last two quadrants.)

(i)



The point P moves in a locus so that the product of its distances from the two points $Q : (-a, 0)$ and $R : (a, 0)$ is constant. Assuming the locus of P goes through the origin, determine the value of the constant, and derive the polar equation of the locus of P .

(Work with the squares of the distances, rather than the distances themselves, and use the law of cosines; the identities $(A+B)(A-B) = A^2 - B^2$ and $\cos 2\theta = 2\cos^2 \theta - 1$ simplify the algebra and produce a simple answer at the end. The resulting curve is a lemniscate,



Condition ① is $|\vec{PQ}| \cdot |\vec{PR}| = \text{const } c$

Condition ② says $P = \text{Origin}$ satisfies above

i.e. $|\vec{OQ}| \cdot |\vec{OR}| = c \Rightarrow c = a^2$

By law of cosines, $|\vec{PQ}|^2 = |\vec{OQ}|^2 + |\vec{OP}|^2 - 2|\vec{OQ}||\vec{OP}| \cos \angle QOP$

$|\vec{PR}|^2 = |\vec{OR}|^2 + |\vec{OP}|^2 - 2|\vec{OR}||\vec{OP}| \cos \angle ROP$

i.e. $a^4 = |\vec{PQ}|^2 |\vec{PR}|^2 = (a^2 + r^2 + 2ar \cos \theta)(a^2 + r^2 - 2ar \cos \theta)$
 $= (a^2 + r^2)^2 - (2ar \cos \theta)^2$

i.e. Simplify to get $2a^2 r^2 (2\cos^2 \theta - 1) = r^4$ i.e. $\boxed{2a^2 \cos 2\theta = r^2}$

For each of the following,

- (i) give the corresponding equation in rectangular coordinates;
 (ii) draw the graph; indicate the direction of increasing θ .

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \tan^{-1} \frac{y}{x} \end{cases}$$

a) $r = \sec \theta$



$$r \cos \theta = 1$$



$$x = 1$$

b) $r = 2a \cos \theta$



$$r^2 = 2a r \cos \theta$$



$$x^2 + y^2 = 2ax$$

c) $r = (a + b \cos \theta)$



$$r^2 = ar + b r \cos \theta$$



$$x^2 + y^2 = a\sqrt{x^2 + y^2} + bx$$

d) $r = a/(b + c \cos \theta)$



$$r(b + c \cos \theta) = a$$

$$\text{i.e. } br + c r \cos \theta = a$$



$$b\sqrt{x^2 + y^2} + cx = a$$

Simplify further

$$b^2(x^2 + y^2) = a^2 - 2acx + c^2x^2$$

$$\text{i.e. } (c^2 - b^2)x^2 - 2acx + a^2 - b^2y^2 = 0$$

The point P moves with constant speed v in the direction of the constant vector $a\mathbf{i} + b\mathbf{j}$. If at time $t = 0$ it is at (x_0, y_0) , what is its position vector function $\mathbf{r}(t)$?

$\vec{OP}(t)$

$a\mathbf{i} + b\mathbf{j}$ $\swarrow$ scalar ≥ 0

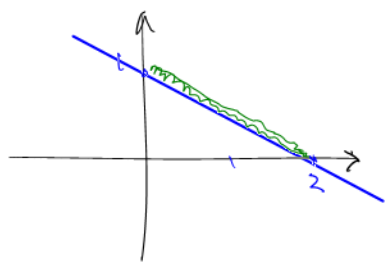
$$\mathbf{r}(t) = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \underbrace{t \cdot v \frac{\begin{pmatrix} a \\ b \end{pmatrix}}{\sqrt{a^2 + b^2}}}_{\text{unit velocity vector}} = \begin{pmatrix} x_0 + \frac{a v t}{\sqrt{a^2 + b^2}} \\ y_0 + \frac{b v t}{\sqrt{a^2 + b^2}} \end{pmatrix}$$

Describe the motions given by each of the following position vector functions, as t goes from $-\infty$ to ∞ . In each case, give the xy -equation of the curve along which P travels, and tell what part of the curve is actually traced out by P .

a) $\mathbf{r} = \underbrace{2\cos^2 t}_{x}\mathbf{i} + \underbrace{\sin^2 t}_{y}\mathbf{j}$

get $x + 2y = 2$

and $\underline{x, y \geq 0}$

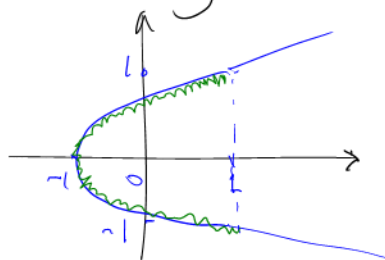


b) $\mathbf{r} = \underbrace{\cos 2t}_{x}\mathbf{i} + \underbrace{\cos t}_{y}\mathbf{j}$

get (double angle formula)

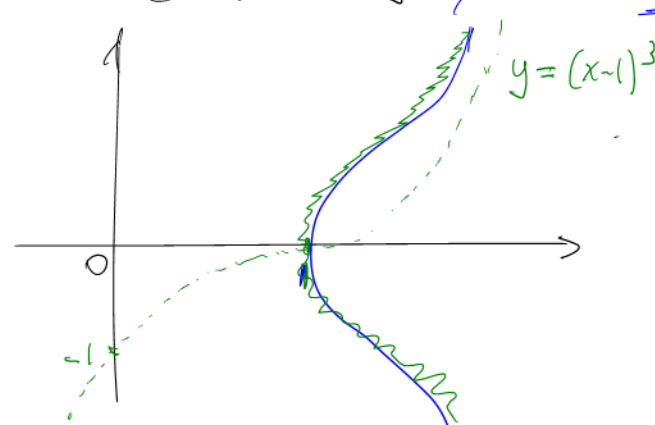
$x = 2y^2 - 1$

and $x, y \in [-1, 1]$



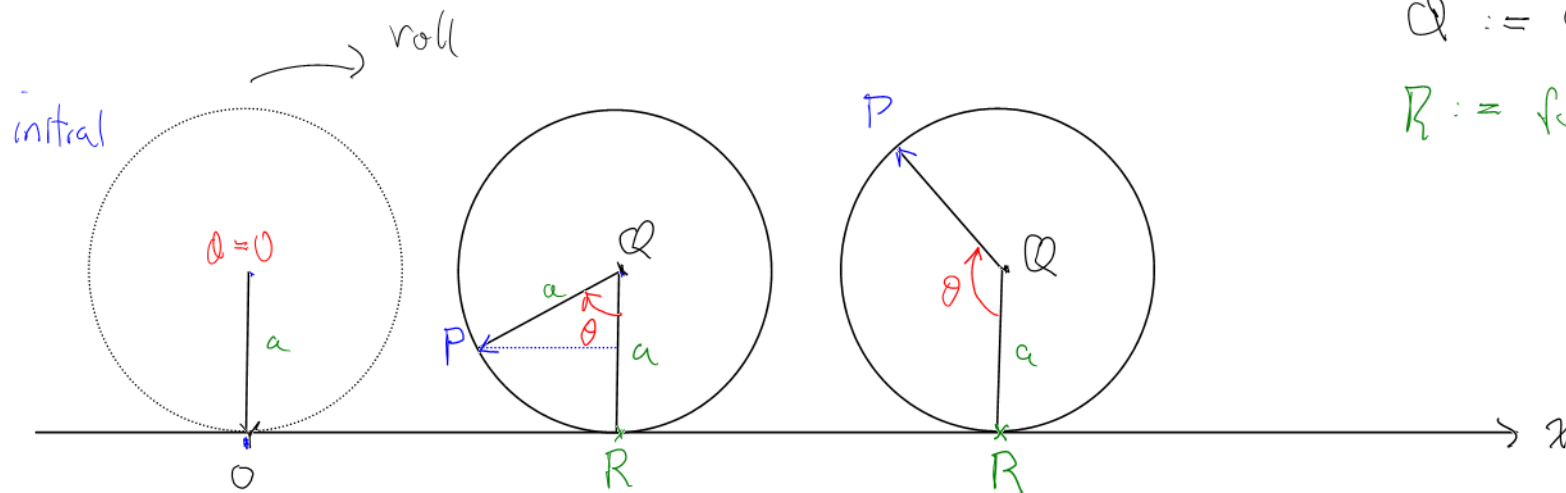
c) $\mathbf{r} = \underbrace{(t^2 + 1)}_{x}\mathbf{i} + \underbrace{t^3}_{y}\mathbf{j}$

get $(x-1)^3 = y^2$ and $\underline{x \geq 1}$



The cycloid is the curve traced out by a fixed point P on a circle of radius a which rolls along the x -axis in the positive direction, starting when P is at the origin O . Find the vector function $\vec{OP}$; use as variable the angle θ through which the circle has rolled.

(Hint: begin by expressing OP as the sum of three simpler vector functions.)



$Q :=$ center of circle

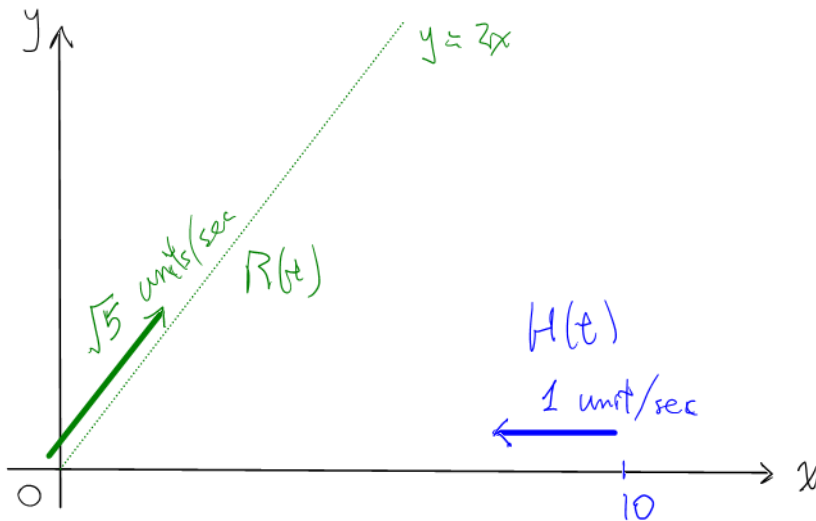
$R :=$ foot of $\perp$ from center of circle to x -axis

$$\begin{aligned} \text{Then } \vec{OP} &= \vec{OR} + \vec{RQ} + \vec{QP} \\ &= \begin{pmatrix} a\theta \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ a \end{pmatrix} + \begin{pmatrix} -a\sin\theta \\ -a\cos\theta \end{pmatrix} = a \cdot \begin{pmatrix} \theta - \sin\theta \\ 1 - \cos\theta \end{pmatrix} \end{aligned}$$

A bow-and-arrow hunter walks toward the origin along the positive x -axis, with unit speed; at time 0 he is at $x = 10$. His arrow (of unit length) is aimed always toward a rabbit hopping with constant velocity $\sqrt{5}$ in the first quadrant along the line $y = 2x$; at time 0 it is at the origin.

a) Write down the vector function $\mathbf{A}(t)$ for the arrow at time t .

b) The hunter shoots (and misses) when closest to the rabbit; when is that?



$$\mathbf{H}(t) = \begin{pmatrix} 10 - 1 \cdot t \\ 0 \end{pmatrix}$$

$$\mathbf{R}(t) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + t \frac{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}{\left| \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right|} \sqrt{5}$$

velocity vector
of rabbit

$$\begin{aligned} \vec{H\mathbf{R}}(t) \quad \cancel{\mathbf{A}(t)} &= \mathbf{R}(t) - \mathbf{H}(t) = t \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 10 - t \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 2t - 10 \\ 2t \end{pmatrix} \end{aligned}$$

$$A(t) = \frac{\vec{HR}(t)}{|\vec{HR}(t)|} = \frac{\begin{pmatrix} 2t-10 \\ 2t \end{pmatrix}}{\sqrt{(2t-10)^2 + (2t)^2}} = \frac{\begin{pmatrix} t-5 \\ t \end{pmatrix}}{\sqrt{2t^2 - 10t + 25}}$$

b) Shoot at time t when $|\vec{HR}(t)|$ is smallest

$$\sqrt{(2t-10)^2 + (2t)^2}$$

Method 1: diff this fn of t , set to 0, solve.

Method 2: find t for which $(2t-10)^2 + (2t)^2$ is smallest

$$= 8t^2 - 40t + 100$$

completing the square $\rightarrow$

$$= 8 \left(\left(t - \frac{5}{2}\right)^2 + \underbrace{\frac{25}{2} - \left(\frac{5}{2}\right)^2}_{\frac{25}{4}} \right)$$

$\Rightarrow$ smallest when $t = \frac{5}{2}$